

Quiz Week 7

Groups



Cyclic groups

1. Indicate whether the statements below is true or false:

- (1) If a group has an element of order 2, and an element of order 3, then it has an element of order 6.
- (2) Every cyclic group of order n (for $n \geq 3$) has at least two generators.
- (3) $\mathbb{Z}_{16}$ has 9 generators.
- (4) If G is a group and $g \in G$ is such that $g^n = 1$, then $\text{o}(g) = n$.

Solution: (1) **False:** $G = S_3$ contains an element of order 2 (for instance) the transposition $(1\ 2)$, and an element of order 3, take the 3-cycle $(1\ 2\ 3)$. But S_3 does not contain any element of order 6, since this would imply S_3 to be cyclic and S_3 is not cyclic.

(2) **True:** Let G be a cyclic group of order $n \geq 3$. Suppose that $g \in G$ is a generator of G . From (a) we know that $\text{o}(g) = \text{o}(g^{-1})$, which implies that g^{-1} is also a generator of G . Notice that $g^{-1} \neq g$; if $g^{-1} = g$, then multiplying by g on both sides we would obtain $g^2 = 1$, that is, $|G| = \text{o}(g) = 2$, but this contradicts the fact that $n \geq 3$. So, g and g^{-1} are two different generators of G .

(3) **False:** We know that the generators of the additive cyclic group $\mathbb{Z}_{16} = \langle \bar{1} \rangle$ are the elements $k\bar{1} = \bar{k} \in \mathbb{Z}_{16}$, where $\gcd(k, 16) = 1$. Hence, $\mathbb{Z}_{16}$ has 8 generators:

$$\bar{1}, \bar{3}, \bar{5}, \bar{7}, \bar{9}, \bar{11}, \bar{13}, \bar{15}.$$

(4) **False:** Take $G = S_3$ again and $g = (1\ 2)$. We have that $g^4 = \varepsilon$ but $\text{o}(g) = 2$.